GENERATE FEATURES

The generate-features module calculates features (image-derived descriptors) from 2D or 3D images and is the first step in the pipeline when used together with the machine-learning modules (model training and inference). Features must be generated for every image used in model training ("labelled") and for the images to be classified ("unlabelled"). In labelled mode, the software retrieves the biological group that the user has assigned to each image, which is then used during training. Unlabelled mode omits this step and is described in 03_inference.

The module can also be run on its own, for example to calculate features for other purposes such as statistical analysis. If you do not follow the image naming convention described below choose "unlabelled" mode.

In addition to generating features, the software offers several simple thresholding methods: two-class Otsu, three-class Otsu, robust (two-class) background, and manual thresholding (which can be optimised in ImageJ). These methods are often sufficient to identify the pixels or voxels belonging to the biofilm, but they do not enable single-cell segmentation. Where single-cell resolution is required, a more sophisticated segmentation technique must be used; the resulting masks can then be imported as TIFF files in MicroICS to calculate the features. The mask needs to have the same name as the original image, with an optional _mask at the end of the name.

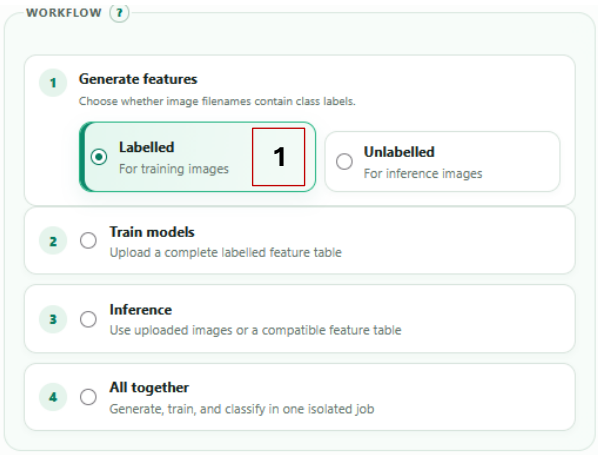
Execution of the MicroICS pipeline relies heavily on the information encoded in image file names, which must follow the scheme below. The biological group used for labelling is parsed directly from these names, so correct naming is essential.

Naming scheme:
<date>_s__st__p_<well_coordinates>_pos<well_position>_tm_<growth_time>_ch__z_
Example:
07112023_s_Lm_st_L1823_p_F02_pos005_tm_24_ch_Syto9_z_21

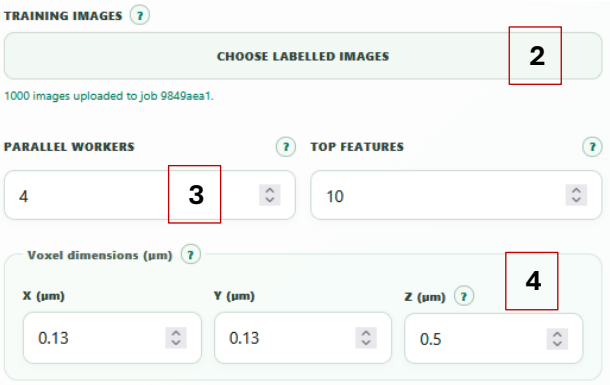
Explanation of each field:
07112023 — date, given as DayMonthYear for easier tracking of results.
s_Lm – s_ indicates the species; Lm denotes Listeria monocytogenes.
st_L1823 — st_ denotes the strain; L1823 is the strain identifier.
p_F02 — p_ denotes the well coordinates on the 96-well plate; F02 is the well.
pos005 — pos denotes the measurement position within that well (here, position 5 in well F02).
tm_24 — tm_ denotes the measurement time; 24 means the images were taken after 24 hours of biofilm growth.
ch_Syto9 — ch_ denotes the channel; Syto9 is the dye/filter used.
z_21 — z_ denotes the number of layers (z-slices) in the 3D image (here, 21).

The fields that are essential for processing are the date, _s_, _st_, _p_, and _pos markers. The names are parsed on the back end according to these exact patterns, and any deviation from them will cause processing to fail or produce incorrect results.

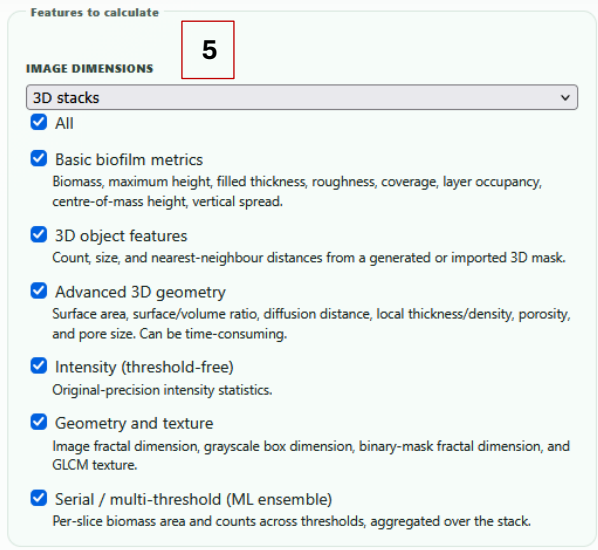
Renaming can be carried out semi-automatically, for example with Advanced Renamer (https://www.advancedrenamer.com/) or with user-defined scripts. Its user guide, including worked examples and illustrations, is available at https://www.advancedrenamer.com/user_guide/v3/complete_guide.



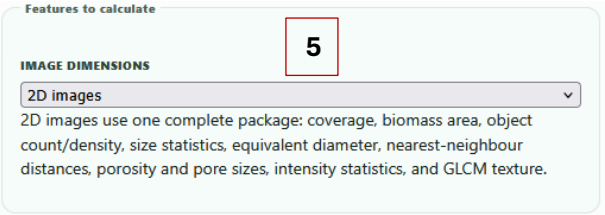
1. Choose “Generate features” in mode “Labelled”.

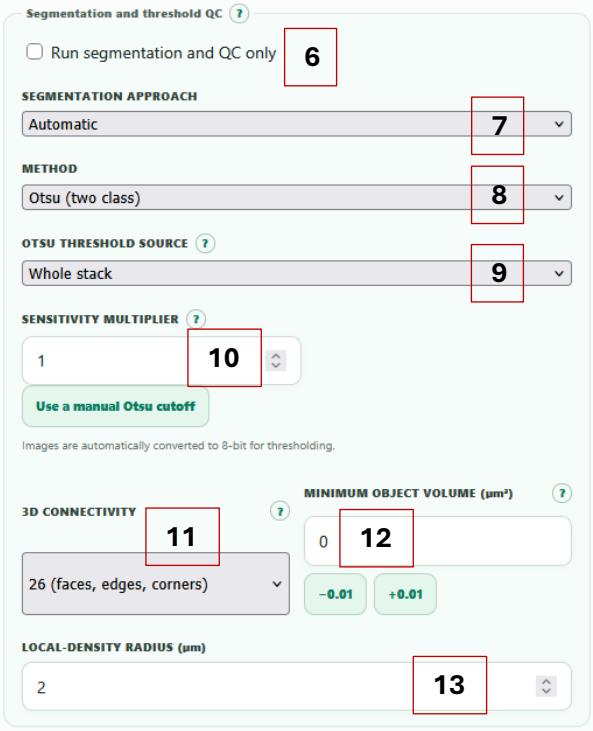


2. Point the software to the folder containing your images in tif format, using “Choose labelled images”.
3. Set the running options. “Parallel workers” – how much of your CPU to dedicate to the analysis: use full capacity when you are not otherwise using the computer, and less if you are working on it at the same time. Suggested workers will provide optimal settings for running.
4. Adjust the pixel/voxel dimensions (the pixel size in micrometres) to match your images.



5. Choose the type of image you will supply to the software (2D or 3D).
For 2D images, a predefined subset of features is calculated automatically. For 3D images, the available feature groups are displayed and can be selected individually (ticked) according to the descriptors you require.





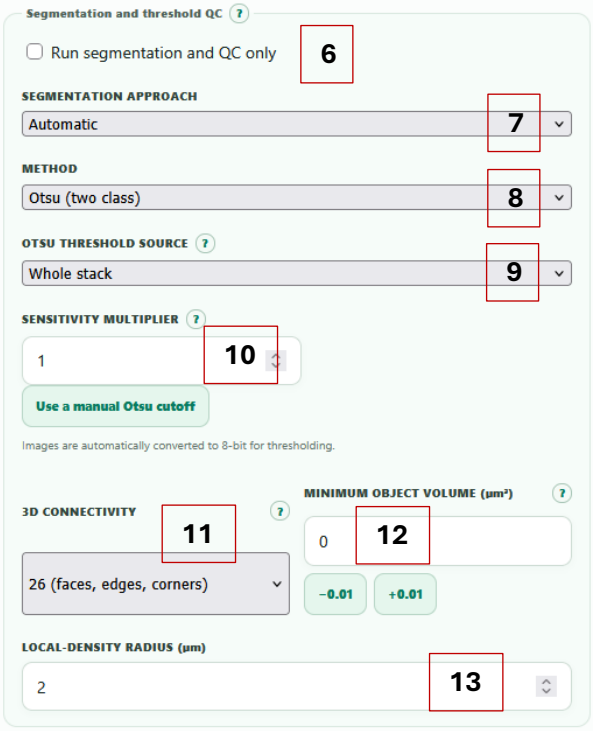
6. (Optional) Run segmentation and QC only – tick this to produce only the segmentation masks, without calculating features. This is useful when testing different algorithms and optimising biomass detection, and is usually done on a representative subset of images.

7. Segmentation approach – choose how the biofilm mask is obtained:

- Automatic – the threshold is computed automatically, using the method selected in step 8.
- Manual – the user supplies a fixed threshold, which can be optimised beforehand in ImageJ. Useful for fine control or when the automatic methods do not give a satisfactory mask.
- Import – load pre-made masks (e.g. ImageJ, from machine learning or single-cell segmentation) as tif files, bypassing the built-in thresholding entirely.

8. Method – the thresholding algorithm used in Automatic mode:

- Otsu (two class) — standard Otsu thresholding, which splits the intensity histogram into two classes (biofilm vs background) by maximizing the between-class variance. A good general default when the biofilm is clearly brighter than the background.
- Otsu (three class) — multi-level Otsu, which splits the histogram into three classes (background / dim / bright). Useful when the image contains two intensity populations (e.g. bright cells and dimmer material); the intermediate “dim” class can be assigned to either biomass or background.
- Robust background (two class) — estimates the background statistically as its mean + 2σ (after trimming the intensity tails) rather than from the full histogram. More robust when the background is noisy or has a baseline offset.



9. Otsu threshold source – the region used to compute the automatic threshold:
   - Whole stack – the threshold is derived from all layers and applied to the whole volume.
   - Substratum – the threshold is derived from the first (substratum) layer. This follows the COMSTAT convention.

10. Sensitivity multiplier – a factor applied to the automatically computed threshold to fine-tune how stringent the segmentation is (default 1).

11. Minimum object volume (µm³) — removes connected objects smaller than this volume (default 0, meaning keep all). Use it to exclude noise; a biologically meaningful value is roughly half a single-cell volume, estimated from cell dimensions measured in ImageJ. Adjust with the −0.01 / +0.01 buttons.

12. 3D connectivity – defines when neighboring voxels belong to the same object: 6 (shared faces only), 18 (faces and edges), or 26 (faces, edges, and corners; default). Higher connectivity merges voxels more readily, giving fewer, larger objects.

13. Local-density radius (µm) — the radius of the sphere used to compute local biomass density around each voxel (default 2 µm). Values are comparable across images only when the same radius is used.

Click “Run pipeline”.

Generate features outputs

Calculated features are provided in the \results in documents with the following names:

- **Datafile.tsv** – produced in labelled mode. This is a single combined table containing the calculated features for all processed images, together with the biological group (Label) parsed from each image name. This file is the input to model training.
- **Unknown.tsv** – produced in unlabelled mode, for images whose biological group is not known. This has the same calculated features but no group label, and serves as the input to inference, where a trained model assigns the predicted group (see 03_inference).

A full description of the calculated features is provided in document 02_descriptions_of_features.

Outputs of segmentation and QC are provided in the \results\feature_generator\segmentation in documents with the following names:

_object_labels_rgb.tif – A colour-coded map of the objects found by connected-component analysis: each separate object is drawn in a different colour. It shows what the algorithm treated as one connected region (e.g. microcolony) versus separate objects (e.g. individual bacteria). Use it to decide how to interpret the object-based metrics (object count, area, nearest neighbour).

_mask.tif – The 3D binary image produced by the thresholding step: voxels recognised as biofilm are foreground (white), and everything else is background (black). This mask is what the software uses to calculate every feature, so it should represent the true biomass as faithfully as possible. With thresholding algorithms that cannot resolve individual bacteria inside aggregates, the mask is always a compromise between capturing all of the biomass as foreground and keeping bacteria as separate objects – resolving single cells in dense regions requires specialised instance-segmentation algorithms. To judge the mask, open the **_overlay.tif**, where the regions recognised as biofilm are shown in transparent red over the original image. The red should trace the real biomass; if it floods into the background or misses biofilm, adjust the threshold or preprocessing steps.

_segmentation.json – A log of the settings used to produce the mask and calculate features for each individual image. This is a plain text file; on Windows, it opens in Notepad.

_histogram.png – A histogram of the 8-bit image (pixel intensity on the x-axis, voxel count on the y-axis), with the applied threshold marked by a dashed red line. The shape of the histogram can help you decide on a thresholding algorithm. For example, for a clear two-peak distribution, a global method like Otsu should perform well. If there is only one dominant background peak with a long tail, a global threshold may not perform optimally; in this case, consider a robust background method, adjusting the threshold manually, or choosing another external segmentation algorithm.

_threshold_curve.png – Shows the detected biomass fraction after applying a series of thresholds. It is useful to judge how sensitive your result is to the threshold and how consistent the replicates are (images from the same conditions).

_thresholding_uint8.tif – The 8-bit (0–255) version of your image. Whatever bit depth you supply (e.g. 12- or 16-bit), the software converts the image to 8-bit for the thresholding step, and this is the image from which the threshold and histogram are computed. Use it to check that the conversion worked correctly and that the signal spans the 0–255 range. (In ImageJ you may need to adjust Brightness/Contrast to see it properly.)

Preparing a mask in ImageJ (for import into MicroICS)

Sometimes ImageJ already provides adequate tools to improve segmentation and produce a usable mask. For very dense biofilms, however, simple thresholding reaches its limits and more advanced segmentation is required. The workflow below covers the ImageJ-based approach; use it when the built-in thresholding is not sufficient and you want to prepare a mask externally and import it (segmentation approach “Import”).

1. Open the stack
Plugins → Bio-Formats → Bio-Formats Importer, and open the 3D TIFF as a hyperstack.
If the image is multi-channel: Image → Color → Split Channels, and keep only the signal channel.
Check that Image → Properties shows the correct voxel size. Bio-Formats usually preserves this; if it is wrong or blank, set it manually.

2. Set the display range, then convert to 8-bit
First use Image → Adjust → Brightness/Contrast to set the display range.
Then apply Image → Type → 8-bit. Conversion maps the current display range to 0–255.

3. Measure the bacterial cell size
Select the line tool, draw along the length of a bacterium, and press Ctrl+M – the length appears in µm.
Measure several cells, and note a typical length and the smallest appearance (end-on or partial cells). Convert these to pixels. Use these values to set the rolling-ball radius in step 4 and the minimum object size in MicroICS.

4. Flatten the background (rolling ball) — only if it improves segmentation
Process → Subtract Background…
Set the rolling-ball radius to ≈ 2 × the measured cell size.
Tick “Stack” / “Process all slices”.

5. Denoise – only if it improves segmentation
If the threshold looks noisy, apply at most a light Gaussian blur ($\sigma \approx 0.5–1$), but skip if damages the quality.

6. Threshold → preview (do not apply yet)
Image → Adjust → Threshold… (Ctrl+Shift+T).
Tick “Stack” (whole-stack histogram) and “Dark background”.
Choose a method (manual, Otsu, or another), and use the same choice for biologically meaningful group of images.
Inspect the red overlay and scroll through several layers; it should trace the biofilm.

7. Save
File → Save As → TIFF, keeping the same dimensions as the original.

Follow the MicroICS naming convention and add _mask at the end of the file name.
This is your mask, ready to import into MicroICS.